

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

- In triangle ABC , if $\sin^3 A + \sin^3 B + \sin^3 C = 3\sin A \cdot \sin B \cdot \sin C$, then the triangle is
 (A) Isosceles (B) right angled
 (C) scalene (D) equilateral
MTS001
- In a triangle ABC , if $\angle C = 105^\circ$, $\angle B = 45^\circ$ and length of side $AC = 2$ units, then the length of the side AB is equal to
 (A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\sqrt{2} + 1$ (D) $\sqrt{3} + 1$
MTS002
- In triangle ABC , if $2b = a + c$ and $A - C = 90^\circ$, then $\sin B$ equals
 [Note: All symbols used have usual meaning in triangle ABC .]
 (A) $\frac{\sqrt{7}}{5}$ (B) $\frac{\sqrt{5}}{8}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{5}}{3}$
MTS005
- Angles A, B and C of a triangle ABC are in A.P. If $\frac{b}{c} = \sqrt{\frac{3}{2}}$, then $\angle A$ is equal to
 [Note: All symbols used have usual meaning in triangle ABC .]
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{2}$
MTS009
- If K is a point on the side BC of an equilateral triangle ABC and if $\angle BAK = 15^\circ$, then the ratio of lengths $\frac{AK}{AB}$ is
 (A) $\frac{3\sqrt{2}(3+\sqrt{3})}{2}$ (B) $\frac{\sqrt{2}(3+\sqrt{3})}{2}$
 (C) $\frac{\sqrt{2}(3-\sqrt{3})}{2}$ (D) $\frac{3\sqrt{2}(3-\sqrt{3})}{2}$
MTS013
- In a triangle $\tan A : \tan B : \tan C = 1 : 2 : 3$, then $a^2 : b^2 : c^2$ equals
 (A) $5 : 8 : 9$ (B) $5 : 8 : 12$
 (C) $3 : 5 : 8$ (D) $5 : 8 : 10$
MTS017

7. In triangle ABC , If $s = 3 + \sqrt{3} + \sqrt{2}$, $3B - C = 30^\circ$, $A + 2B = 120^\circ$, then the length of longest side of triangle is

[Note: All symbols used have usual meaning in triangle ABC .]

- (A) 2 (B) $2\sqrt{2}$
(C) $2(\sqrt{3} + 1)$ (D) $\sqrt{3} - 1$

MTS020

8. If in a triangle $\sin A : \sin C = \sin(A - B) : \sin(B - C)$, then a^2, b^2, c^2

[Note: All symbols used have usual meaning in triangle ABC .]

- (A) are in A.P. (B) are in G.P.
(C) are in H.P. (D) are in A.G.P

MTS016

9. In an acute triangle ABC , $\angle ABC = 45^\circ$, $AB = 3$ and $AC = \sqrt{6}$. The angle $\angle BAC$, is

- (A) 60° (B) 65° (C) 75° (D) 85°

MTS018

10. In triangle ABC , if $AC = 8$, $BC = 7$ and D lies between A and B such that $AD = 2$, $BD = 4$, then the length CD equals

- (A) $\sqrt{46}$ (B) $\sqrt{48}$ (C) $\sqrt{51}$ (D) $\sqrt{75}$

MTS004

11. In a triangle ABC , $a^3 + b^3 + c^3 = c^2(a + b + c)$

(All symbol used have usual meaning in a triangle.)

Statement-1: The value of $\angle C = 60^\circ$.

Statement -2: $\triangle ABC$ must be equilateral.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

MTS006

12. In triangle ABC , If $\frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$ then angle C is equal to

[Note: All symbols used have usual meaning in triangle ABC .]

- (A) 30° (B) 45°
(C) 60° (D) 90°

MTS007

13. The ratio of the sides of a triangle ABC is $1:\sqrt{3}:2$. Then ratio of $\angle A : \angle B : \angle C$ is

(A) $3 : 5 : 2$

(B) $1 : \sqrt{3} : 2$

(C) $3 : 2 : 1$

(D) $1 : 2 : 3$

MTS015

14. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. If $a = 1, b = 3$ and $C = 60^\circ$, then $\sin^2 B$ is equal to

[Note: All symbols used have usual meaning in triangle ABC .]

(A) $\frac{27}{28}$

(B) $\frac{3}{28}$

(C) $\frac{81}{28}$

(D) $\frac{1}{3}$

MTS019

15. In a triangle ABC , if $b = (\sqrt{3} - 1)a$ and $\angle C = 30^\circ$, then the value of $(A - B)$ is equal to

(All symbols used have usual meaning in a triangle.)

(A) 30°

(B) 45°

(C) 60°

(D) 75°

MTS008

16. In a triangle ABC , if $a = 13, b = 14$ and $c = 15$, then angle A is equal to

(All symbols used have their usual meaning in a triangle.)

(A) $\sin^{-1} \frac{4}{5}$

(B) $\sin^{-1} \frac{3}{5}$

(C) $\sin^{-1} \frac{3}{4}$

(D) $\sin^{-1} \frac{2}{3}$

MTS003

17. AD and BE are the medians of a triangle ABC . If $AD = 4, \angle DAB = \frac{\pi}{6}, \angle ABE = \frac{\pi}{3}$, then area of triangle ABC equals

(A) $\frac{8}{3}$

(B) $\frac{16}{3}$

(C) $\frac{32}{3}$

(D) $\frac{32}{9}\sqrt{3}$

MTS010

18. In a triangle ABC , if $(a + b + c)(a + b - c)(b + c - a)(c + a - b) = \frac{8a^2b^2c^2}{a^2 + b^2 + c^2}$, then the triangle is

[Note: All symbols used have usual meaning in triangle ABC .]

(A) isosceles

(B) right angled

(C) equilateral

(D) scalene

MTS011

19. For right angled isosceles triangle, $\frac{r}{R} =$

[Note: All symbols used have usual meaning in triangle ABC .]

- (A) $\tan \frac{\pi}{12}$ (B) $\cot \frac{\pi}{12}$ (C) $\tan \frac{\pi}{8}$ (D) $\cot \frac{\pi}{8}$

MTS012

20. Let ABC be a right triangle with length of side $AB = 3$ and hypotenuse $AC = 5$.

If D is a point on BC such that $\frac{BD}{DC} = \frac{AB}{AC}$, then AD is equal to

- (A) $\frac{4\sqrt{3}}{3}$ (B) $\frac{3\sqrt{5}}{2}$ (C) $\frac{4\sqrt{5}}{3}$ (D) $\frac{5\sqrt{3}}{4}$

MTS014

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

- Given an acute triangle ABC such that $\sin C = \frac{4}{5}$, $\tan A = \frac{24}{7}$ and $AB = 50$. Then-
 (A) centroid, orthocentre and incentre of ΔABC are collinear
 (B) $\sin B = \frac{4}{5}$
 (C) $\sin B = \frac{4}{7}$
 (D) area of $\Delta ABC = 1200$
MTS036
- In ΔABC , angle A is 120° , $BC + CA = 20$ and $AB + BC = 21$, then
 (A) $AB > AC$ (B) $AB < AC$
 (C) ΔABC is isosceles (D) area of $\Delta ABC = 14\sqrt{3}$
MTS037
- In which of the following situations, it is possible to have a triangle ABC ?
 (All symbols used have usual meaning in a triangle.)
 (A) $(a + c - b)(a - c + b) = 4bc$
 (B) $b^2 \sin 2C + c^2 \sin 2B = ab$
 (C) $a = 3, b = 5, c = 7$ and $C = \frac{2\pi}{3}$
 (D) $\cos\left(\frac{A-C}{2}\right) = \cos\left(\frac{A+C}{2}\right)$
MTS038
- In ΔABC , angle A, B and C are in the ratio $1 : 2 : 3$, then which of the following is (are) correct?
 (All symbol used have usual meaning in a triangle.)
 (A) Circumradius of $\Delta ABC = c$ (B) $a : b : c = 1 : \sqrt{3} : 2$
 (C) Perimeter of $\Delta ABC = 3 + \sqrt{3}$ (D) Area of $\Delta ABC = \frac{\sqrt{3}}{8} c^2$
MTS040
- In a triangle ABC , let $BC = 1$, $AC = 2$ and measure of angle C is 30° . Which of the following statement(s) is (are) correct?
 (A) $2\sin A = \sin B$
 (B) Length of side AB equals $5 - 2\sqrt{3}$
 (C) Measure of angle A is less than 30°
 (D) Circumradius of triangle ABC is equal to length of side AB
MTS041
- If the lengths of the medians AD, BE and CF of triangle ABC are 6, 8, 10 respectively, then-
 (A) AD & BE are perpendicular
 (B) BE and CF are perpendicular
 (C) area of $\Delta ABC = 32$
 (D) area of $\Delta DEF = 8$
MTS039

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 7 to 8

Let incircle of $\triangle ABC$ with $\angle B = \angle C$, touches AB, BC, CA at R, P, Q respectively. If inradius ' r ' is integer and perimeter of $\triangle ABC$ is smallest possible integer such that $\frac{2}{AR} + \frac{5}{BP} + \frac{5}{CQ} = \frac{6}{r}$ then

7. The value of '
- r
- ' is

(A) 4 (B) 3 (C) 2 (D) 1

MTS042

8. Area of
- $\triangle ABC$
- is

(A) 15 sq. unit (B) 21 sq. unit (C) 24 sq. unit (D) 27 sq. unit

MTS043

MATRIX MATCH TYPE QUESTION

9. If symbols have usual notations in triangle
- ABC
- , then match the following columns:

Column-I

- (A) $r r_1 r_2 r_3$
 (B) $r_1 r_2 r_3$
 (C) $r_1 + r_2 - r_3 + r$
 (D) $\frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab}$

Column-II

- (P) rs^2
 (Q) $4R \cos C$
 (R) $\frac{1}{2Rr}$
 (S) Δ^2

MTS044

10. If sides of a triangle
- ABC
- are
- $a = 6, b = 8$
- &
- $c = 10$
- , then match the following (using usual notations)

Column-I

- (A) r is equal to
 (B) r_1 is equal to
 (C) R is equal to
 (D) Δ is equal to

Column-II

- (P) 4
 (Q) 24
 (R) 2
 (S) 6
 (T) 5

MTS067

EXERCISE (S)

1. In $\triangle ABC$, if $a = 2b$ and $A = 3B$, then the value of $\frac{2c^2}{b^2}$ is equal to
[Note: All symbols used have usual meaning in triangle ABC .]
MTS024
2. If the angle A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of expression $E = \left(\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A \right)$, is λ , then the value of λ^2 is
MTS025
3. The sides a, b, c (taken in that order) of triangle ABC are in A.P.
If $\cos \alpha = \frac{a}{b+c}, \cos \beta = \frac{b}{c+a}, \cos \gamma = \frac{c}{a+b}$ then $\tan^2 \left(\frac{\alpha}{2} \right) + \tan^2 \left(\frac{\gamma}{2} \right)$ is equal to $\frac{k}{6}$, then the value of k is
[Note: All symbols used have usual meaning in triangle ABC .]
MTS023
4. In $\triangle ABC$ with usual notations, if $a = 3, b = 2$ and $\cos(A - B) = \frac{12}{13}$, then area of $\triangle ABC$ is
MTS047
5. If in a $\triangle ABC$, $a = 6, b = 3$ and $\cos(A - B) = 4/5$, then find its area.
[Note: All symbols used have usual meaning in triangle ABC .]
MTS065
6. A triangle has vertices A, B and C , and the respective opposite sides have lengths a, b and c . This triangle is inscribed in a circle of radius R . If $b = c = 1$ and the altitude from A to side BC has length $\sqrt{\frac{2}{3}}$, then $8R^2$ equals
[Note: All symbols used have usual meaning in triangle ABC .]
MTS021
7. A circle is inscribed in a right triangle ABC , right angled at C . The circle is tangent to the segment AB at D and length of segments AD and DB are 7 and 13 respectively. Area of triangle ABC is equal to
MTS022
8. In a $\triangle ABC$ with usual notations, if $3a + 3b = 4c$ then $\cot \frac{A}{2} \cot \frac{B}{2}$ is
MTS045
9. In $\triangle ABC$ (with usual notations), if $r_1 = \Delta$ & $r_1 = r + r_2 + r_3$, then r is equal to
MTS046
10. If the area of a triangle is 100 sq.cm, $r_1 = 10$ cm and $r_2 = 50$ cm, then find the value of $(b - a)$.
[Note: All symbols used have usual meaning in triangle ABC .]
MTS066

EXERCISE - JEE (Main) PYQ

1. If $5, 5r, 5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to : [JEE (Main) 2019]
 (1) $\frac{3}{2}$ (2) $\frac{3}{4}$ (3) $\frac{5}{4}$ (4) $\frac{7}{4}$ MTS026
2. With the usual notation, in $\triangle ABC$, if $\angle A + \angle B = 120^\circ$, $a = \sqrt{3} + 1$ and $b = \sqrt{3} - 1$, then the ratio $\angle A : \angle B$, is : [JEE (Main) 2019]
 (1) $7 : 1$ (2) $5 : 3$ (3) $9 : 7$ (4) $3 : 1$ MTS027
3. In a triangle, the sum of lengths of two sides is x and the product of the lengths of the same two sides is y . If $x^2 - c^2 = y$, where c is the length of the third side of the triangle, then the circumradius of the triangle is : [JEE (Main) 2019]
 (1) $\frac{y}{\sqrt{3}}$ (2) $\frac{c}{\sqrt{3}}$ (3) $\frac{c}{3}$ (4) $\frac{3}{2}y$ MTS028
4. Given $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$ for a $\triangle ABC$ with usual notation. If $\frac{\cos A}{\alpha} = \frac{\cos B}{\beta} = \frac{\cos C}{\gamma}$, then the ordered triad (α, β, γ) has a value :- [JEE (Main) 2019]
 (1) $(3, 4, 5)$ (2) $(19, 7, 25)$ (3) $(7, 19, 25)$ (4) $(5, 12, 13)$ MTS029
5. If the lengths of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then a ratio of lengths of the sides of this triangle is : [JEE (Main) 2019]
 (1) $5 : 9 : 13$ (2) $5 : 6 : 7$ (3) $4 : 5 : 6$ (4) $3 : 4 : 5$ MTS030
6. The angles A, B and C of a triangle ABC are in A.P. and $a : b = 1 : \sqrt{3}$. If $c = 4$ cm, then the area (in sq. cm) of this triangle is : [JEE (Main) 2019]
 (1) $4\sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$ (3) $2\sqrt{3}$ (4) $\frac{4}{\sqrt{3}}$ MTS031
7. If a $\triangle ABC$ has vertices $A(-1, 7), B(-7, 1)$ and $C(5, -5)$, then its orthocentre has coordinates: [JEE (Main) 2020]
 (1) $(3, -3)$ (2) $\left(-\frac{3}{5}, \frac{3}{5}\right)$ (3) $(-3, 3)$ (4) $\left(\frac{3}{5}, -\frac{3}{5}\right)$ MTS032
8. If in a triangle ABC , $AB = 5$ units, $\angle B = \cos^{-1}\left(\frac{3}{5}\right)$ and radius of circumcircle of $\triangle ABC$ is 5 units, then the area (in sq. units) of $\triangle ABC$ is : [JEE (Main) 2021]
 (1) $10 + 6\sqrt{2}$ (2) $8 + 2\sqrt{2}$ (3) $6 + 8\sqrt{3}$ (4) $4 + 2\sqrt{3}$ MTS033

9. In $\triangle ABC$, the lengths of sides AC and AB are 12 cm and 5 cm , respectively. If the area of $\triangle ABC$ is 30 cm^2 and R and r are respectively the radii of circumcircle and incircle of $\triangle ABC$, then the value of $2R + r$ (in cm) is equal to _____.
[JEE (Main) 2021] MTS034
10. The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for $x = k$, the area of the triangle is maximum, then $3k^2$ is equal to :
(1) 5 (2) 8 (3) 10 (4) 12
[JEE (Main) 2022] MTS035
11. In a triangle ABC , if $\cos A + 2 \cos B + \cos C = 2$ and the lengths of the sides opposite to the angles A and C are 3 and 7 respectively, then $\cos A - \cos C$ is equal to
(1) $\frac{3}{7}$ (2) $\frac{9}{7}$ (3) $\frac{10}{7}$ (4) $\frac{5}{7}$
[JEE (Main) 2023] MTS064
12. In a triangle ABC , $BC = 7$, $AC = 8$, $AB = \alpha \in \mathbb{N}$ and $\cos A = \frac{2}{3}$. If $49\cos(3C) + 42 = \frac{m}{n}$, where $\gcd(m, n) = 1$, then $m + n$ is equal to _____.
[JEE (Main) 2024] MTS116

EXERCISE - JEE (Advanced) PYQ

1. Let ABC and ABC' be two non-congruent triangles with sides $AB = 4$, $AC = AC' = 2\sqrt{2}$ and angle $B = 30^\circ$. The absolute value of the difference between the areas of these triangles is
[JEE (Advanced) 2009]
MTS048
2. If the angle A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the length of the sides opposite to A, B and C respectively, then the value of the expression $\frac{a}{c}\sin 2C + \frac{c}{a}\sin 2A$, is -
[JEE (Advanced) 2010]
(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$
MTS049
3. Consider a triangle ABC and let a, b and c denote the length of the sides opposite to vertices A, B and C respectively. Suppose $a = 6, b = 10$ and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to
[JEE (Advanced) 2010]
MTS050
4. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C respectively. The value(s) of x for which $a = x^2 + x + 1, b = x^2 - 1$ and $c = 2x + 1$ is/are
[JEE (Advanced) 2010]
(A) $-(2+\sqrt{3})$ (B) $1+\sqrt{3}$ (C) $2+\sqrt{3}$ (D) $4\sqrt{3}$
MTS051
5. Let PQR be a triangle of area Δ with $a = 2, b = \frac{7}{2}$ and $c = \frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals
[JEE (Advanced) 2012]
(A) $\frac{3}{4\Delta}$ (B) $\frac{45}{4\Delta}$ (C) $\left(\frac{3}{4\Delta}\right)^2$ (D) $\left(\frac{45}{4\Delta}\right)^2$
MTS052
6. In a triangle PQR , P is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides PQ, QR and RP at N, L and M respectively, such that the lengths of PN, QL and RM are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are)
[JEE (Advanced) 2013]
(A) 16 (B) 18 (C) 24 (D) 22
MTS053

7. In a triangle the sum of two sides is x and the product of the same two sides is y . If $x^2 - c^2 = y$, where c is a third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is - [JEE (Advanced) 2014]

(A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$ (C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$

MTS054

8. In a triangle XYZ , let x, y, z be the lengths of sides opposite to the angles X, Y, Z , respectively and $2s = x + y + z$. If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then- [JEE (Advanced) 2016]

- (A) area of the triangle XYZ is $6\sqrt{6}$
 (B) the radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$
 (C) $\sin \frac{X}{2} \sin \frac{Y}{2} \sin \frac{Z}{2} = \frac{4}{35}$
 (D) $\sin^2 \left(\frac{X+Y}{2} \right) = \frac{3}{5}$

MTS055

9. In a triangle PQR , let $\angle PQR = 30^\circ$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10, respectively. Then, which of the following statement(s) is (are) TRUE ? [JEE (Advanced) 2018]

- (A) $\angle QPR = 45^\circ$
 (B) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120^\circ$
 (C) The radius of the incircle of the triangle PQR is $10\sqrt{3} - 15$
 (D) The area of the circumcircle of the triangle PQR is 100π .

MTS056

10. In a non-right-angled triangle ΔPQR , let p, q, r denote the lengths of the sides opposite to the angles at P, Q, R respectively. The median from R meets the side PQ at S , the perpendicular from P meets the side QR at E , and RS and PE intersect at O . If $p = \sqrt{3}, q = 1$, and the radius of the circumcircle of the ΔPQR equals 1, then which of the following options is/are correct ? [JEE (Advanced) 2019]

- (A) Area of $\Delta SOE = \frac{\sqrt{3}}{12}$ (B) Radius of incircle of $\Delta PQR = \frac{\sqrt{3}}{2}(2 - \sqrt{3})$
 (C) Length of $RS = \frac{\sqrt{7}}{2}$ (D) Length of $OE = \frac{1}{6}$

MTS057

11. Let x, y and z be positive real numbers. Suppose x, y and z are lengths of the sides of a triangle opposite to its angles X, Y and Z , respectively. If $\tan \frac{X}{2} + \tan \frac{Z}{2} = \frac{2y}{x+y+z}$, then which of the following statements is/are TRUE? [JEE (Advanced) 2020]

- (A) $2Y = X + Z$ (B) $Y = X + Z$
 (C) $\tan \frac{X}{2} = \frac{x}{y+z}$ (D) $x^2 + z^2 - y^2 = xz$

MTS058

Solution of Triangles

12. In a triangle ABC , let $AB = \sqrt{23}$, $BC = 3$ and $CA = 4$. Then the value of $\frac{\cot A + \cot C}{\cot B}$ is _____.
[JEE (Advanced) 2021] MTS059
13. Consider a triangle PQR having sides of lengths p, q and r opposite to the angles P, Q and R , respectively. Then which of the following statements is (are) TRUE? [JEE (Advanced) 2021]
- (A) $\cos P \geq 1 - \frac{p^2}{2qr}$
- (B) $\cos R \geq \left(\frac{q-r}{p+q}\right) \cos P + \left(\frac{p-r}{p+q}\right) \cos Q$
- (C) $\frac{q+r}{p} < 2 \frac{\sqrt{\sin Q \sin R}}{\sin P}$
- (D) If $p < q$ and $p < r$, then $\cos Q > \frac{p}{r}$ and $\cos R > \frac{p}{q}$
14. Let $PQRS$ be a quadrilateral in a plane, where $QR = 1$, $\angle PQR = \angle QRS = 70^\circ$, $\angle PQS = 15^\circ$ and $\angle PRS = 40^\circ$. If $\angle RPS = \theta^\circ$, $PQ = \alpha$ and $PS = \beta$, then the interval(s) that contain(s) the value of $4\alpha\beta\sin\theta^\circ$ is/are [JEE (Advanced) 2022]
- (A) $(0, \sqrt{2})$ (B) $(1, 2)$ (C) $(\sqrt{2}, 3)$ (D) $(2\sqrt{2}, 3\sqrt{2})$

MTS061

PARAGRAPH "I"

Consider an obtuse angled triangle ABC in which the difference between the largest and the smallest angle is $\frac{\pi}{2}$ and whose sides are in arithmetic progression. Suppose that the vertices of this triangle lie on a circle of radius 1.

(There are two questions based on PARAGRAPH "I", the question given below is one of them)

15. Let a be the area of the triangle ABC . Then the value of $(64a)^2$ is [JEE (Advanced) 2023] MTS062
16. Then the inradius of the triangle ABC is [JEE (Advanced) 2023] MTS063

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	D	C	C	C	A	C	A	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	D	A	C	A	D	B	C	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	
Ans.	A,B,D	A,D	B,C	B,D	A,C,D	A,C,D	C	D	
Que.	9			10					
Ans.	$A \rightarrow S; B \rightarrow P; C \rightarrow Q; D \rightarrow R$			$A \rightarrow R; B \rightarrow P; C \rightarrow T; D \rightarrow Q$					

EXERCISE (S)

1.	6	2.	3	3.	4	4.	3	5.	9
6.	3	7.	91	8.	7	9.	1	10.	8

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	2	3	3	3	3	3	15	3
Que.	11	12								
Ans.	3	39								

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	D	3	B	C	B,D	B	A,C,D	B,C,D	B,C,D
Que.	11	12	13	14	15	16				
Ans.	B,C	2	A,B	A,B	1007.99 To 1008.01	0.24 to 0.26				